GEOMETRY-AWARE RAG: ENHANCING RETRIEVAL THROUGH MIXED-CURVATURE PRODUCT MANIFOLDS

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ABSTRACT

Dense retrieval is a core component of RAG systems, but standard retrievers usually represent both queries and documents in a single Euclidean embedding space. This can be limiting when the corpus contains hierarchical or mixed structural relationships. We propose **Manifold-RAG**, a geometry-aware dense retrieval framework that projects frozen SBERT embeddings into a mixed-curvature *product manifold* $\mathbb{R}^{d_1} \times \mathbb{B}^{d_2} \times \mathbb{S}^{d_3}$ combining Euclidean, hyperbolic Poincaré-ball, and spherical components. We systematically evaluate on three datasets spanning flat (FiQA), mildly hierarchical (SciFact), and explicitly hierarchical (a custom Wikipedia corpus with hierarchy-aware QA) structure. Surprisingly, we find that Euclidean projection heads remain strongest overall, while purely hyperbolic projections underperform across all datasets. However, mixed-curvature product spaces improve structure-sensitive hierarchy-disambiguation queries (+9.1 NDCG@10 points over Euclidean) and achieve the best trained-head performance on FiQA at dim$\geq$32. Dimension sweep, component ablation, and query-adaptive weight experiments further characterize when and why mixed-curvature geometry helps, revealing that the practical benefit is task-dependent rather than universal.

1 INTRODUCTION

Retrieval-Augmented Generation (RAG) systems ground LLM outputs by retrieving relevant documents via vector similarity search. Conventional dense retrieval embeds queries and documents into Euclidean space and ranks by cosine similarity. While effective, Euclidean embeddings cannot faithfully represent data with non-Euclidean structure—hierarchies, cycles, or multi-scale organization—without significant distortion (Nickel & Kiela, 2017; Sala et al., 2018).

We propose Manifold-RAG, which projects embeddings into a mixed-curvature product manifold: $\mathbb{R}^{d_1} \times \mathbb{B}^{d_2} \times \mathbb{S}^{d_3}$ (Euclidean, hyperbolic Poincaré ball, and spherical components). Each captures different structure: hyperbolic for tree-like ontologies, spherical for mutually exclusive categories, Euclidean for flat similarity.

Motivation. We validate this intuition with a synthetic experiment: embedding a balanced tree (depth=6, branching=3, 729 leaves) into Euclidean vs. Poincaré space by optimizing embedding distances to match true tree distances. Results show Poincaré achieves dramatically lower distortion, especially at low dimensions (10$\times$ lower at d=4), confirming that hyperbolic geometry is better suited for hierarchical data.

Table 1: Tree embedding distortion (lower is better).

Dim	Euclidean	Poincaré	Ratio
2	12740	2828	4.5×
4	242	24	10.1×
8	90	10	9.1×
16	26	8	3.3×

The gap narrows at higher dimensions, suggesting hyperbolic geometry's practical value lies in *dimension efficiency*—achieving equivalent fidelity with fewer parameters. Our project investigates whether this advantage transfers to real retrieval tasks.

Our study asks three questions:

1. Can non-Euclidean projection heads improve dense retrieval over Euclidean projection heads?
2. Does hyperbolic geometry help more in low-dimensional or hierarchical retrieval settings?
3. When does mixed-curvature geometry help or hurt?

Our contributions are:

1. We propose Manifold-RAG, which projects frozen SBERT embeddings into Euclidean, Poincaré, and mixed-curvature product manifolds for dense retrieval.
2. We conduct systematic comparisons on SciFact, FiQA, and a custom Wiki hierarchy QA dataset, including dimension sweeps, component ablations, and adaptive weight analysis.
3. We find that Euclidean projections remain strongest overall, but product manifolds provide a clear advantage on hierarchy disambiguation, demonstrating that the benefit of geometry-aware retrieval is task-dependent rather than universal.

2 Related Work

Non-Euclidean & Product Manifolds. Hyperbolic spaces excel at embedding hierarchical trees with minimal distortion, based on the fact that trees can be embedded into 2D hyperbolic space with arbitrarily low distortion (Sarkar, 2011; Nickel & Kiela, 2017). Furthermore, mixed-curvature product manifolds combine Euclidean, hyperbolic, and spherical factors to better capture diverse, heterogeneous geometries (Gu et al., 2019). Although these geometric priors have advanced word embeddings (Tifrea et al., 2019) and generalized neural network primitives (Ganea et al., 2018), their application to dense retrieval remains unexplored. We adapt these non-Euclidean geometries to capture the complex, multi-scale structural relationships inherent in RAG corpora.

Dense Retrieval and RAG. Modern RAG systems (Lewis et al., 2020; Guu et al., 2020) and dense retrievers operate almost exclusively in Euclidean space, relying on pretrained transformers for nearest-neighbor retrieval (Karpukhin et al., 2020; Izacard et al., 2022), multi-vector late interaction (Khattab & Zaharia, 2020) or contrastive training objectives (Xiong et al., 2021). Despite their advantages, this flat geometry underlying limits the representation of structural knowledge. Our Product starts from this by projecting embeddings into a product manifold, effectively reducing hallucinations and improving the retrieval component for knowledge-intensive tasks.

Benchmarks & Optimization. We evaluated our mixed-curvature representations in SciFact and FiQA from BEIR (Thakur et al., 2021), supplemented by a custom Wikipedia hierarchy corpus to specifically test structure-sensitive retrieval. Training on non-Euclidean manifolds requires Primannian gradient methods. Thus, we utilize Riemannian Adam (Becigneul & Ganea, 2019) via the geoopt library to optimize projection heads through tangent-space and retraction updates.

3 Method Setup

In this section, we present the Manifold-RAG framework, starting with its high-level architecture (Figure 1) and proceeding to its core geometric components, implementation details, and theoretical foundations. Given a query or document text, we first encode it using a frozen SBERT encoder. The resulting embedding is then mapped into multiple geometric components, including Euclidean, Poincaré, and spherical spaces. Retrieval is performed by ranking documents according to a weighted product score computed across these components.

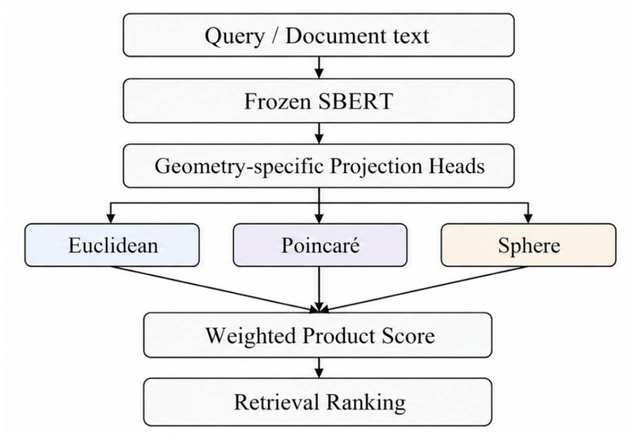

Figure 1: Overview of Manifold-RAG. Query and document texts are encoded by a frozen SBERT encoder and mapped into Euclidean, Poincaré, and spherical components through linear projection heads. Retrieval is performed by ranking documents according to a weighted product score across the three geometries.

3.1 Core Idea

Standard dense retrieval projects queries and documents into a single Euclidean space and ranks by cosine similarity. This implicitly assumes that all semantic relationships are "flat." However, knowledge corpora contain a heterogeneous structure: hierarchical taxonomies (e.g., Wikipedia categories), mutually exclusive groupings (e.g., scientific disciplines) and flat topical similarity. We propose replacing the Euclidean projection with a *product manifold* projection that decomposes the embedding space into three geometric components, each suited to a different type of structure:

- **Euclidean component** $\mathbb{R}^{d_1}$: captures flat semantic similarity via L_2-normalized cosine distance.
- **Hyperbolic component** $\mathbb{B}^{d_2}$ (Poincaré ball, curvature c=1): captures tree-like hierarchical relationships. Points near the origin represent abstract concepts; points near the boundary represent specific ones.
- **Spherical component** $\mathbb{S}^{d_3}$: captures mutually exclusive categories via geodesic (arc-cosine) distance on the unit sphere.

The combined ranking score in the product space is a learnable weighted sum of squared component distances:

$$D_{\text{prod}}(x, y) = \alpha^2\, d_{\mathbb{R}}^2(x_e, y_e) \;+\; \beta^2\, d_{\mathbb{B}}^2(x_h, y_h) \;+\; \gamma^2\, d_{\mathbb{S}}^2(x_s, y_s) \tag{1}$$

where α, β, γ are learnable scalar weights (initialized to 1.0) and $d_{\mathbb{R}}$, $d_{\mathbb{B}}$, $d_{\mathbb{S}}$ are the native distances in each component space. Retrieval ranks documents by ascending D_{prod}. Note that D_{prod} is a weighted squared distance (not a metric in the strict sense) used as a ranking score; this avoids the square root for computational simplicity and is monotonic with respect to the true product metric.

3.2 Implementation Details

Projection Heads. We train lightweight projection heads on top of frozen SBERT (`all-MiniLM-L6-v2`, 384d) embeddings. Each head is a single linear layer $W \in \mathbb{R}^{384\times d}$ that maps the SBERT output to the target manifold:

- **Euclidean Head**: $f(x) = \frac{Wx}{\|Wx\|}$ (project + L_2 normalize).
- **Poincaré Head**: $f(x) = \exp_0^c(s \cdot Wx)$ where $\exp_0^c$ is the exponential map at the origin of the Poincaré ball with curvature c=1, and s is a learnable tangent-space scale (initialized to 0.3) that controls how close to the boundary points are mapped.
- **Product Head**: three parallel projections W_e, W_h, W_s producing the Euclidean (L_2-normalized), hyperbolic ($\exp_0$), and spherical (L_2-normalized) components respectively.

Training. All heads are trained with triplet margin loss (margin$= 0.5$):

$$\mathcal{L} = \frac{1}{|B|} \sum_{(q,p,N)} \left[d(q,p) - \min_{n \in N} d(q,n) + 0.5 \right]_+$$

using Riemannian Adam (Becigneul & Ganea, 2019) via `geoopt`. We sample 7 negatives per query: 4 hard negatives from an index of same-section-title documents across different hierarchy branches (when available), plus 3 random negatives.

Warm-Starting. The Euclidean head is trained first (20 epochs, $lr = 10^{-3}$). Its weights are then copied to initialize the Poincaré head (50 epochs, $lr = 10^{-4}$) and the Product head (30 epochs, $lr = 10^{-3}$), with the Euclidean projection weights partitioned across the three sub-projections.

Hierarchy Norm Regularization. For the Wiki dataset, we add a regularization term to Poincaré and Product heads that encourages the hyperbolic component norms to reflect hierarchy level:

$$\mathcal{L}_{\text{reg}} = \lambda \sum_i (\|h_i\| - t_i)^2, \quad t_i \in \{0.20, 0.45, 0.70\}$$

where t_i maps level-1 categories to 0.20 (near origin), level-2 articles to 0.45, and level$\geq$3 sections to 0.70 (near boundary). We set $\lambda = 0.2$.

Adaptive Product Head. We additionally experiment with a query-adaptive variant where the distance weights (α, β, γ) are predicted per-query by a small MLP: $[\alpha_q, \beta_q, \gamma_q] = \text{Softplus}(W_2 \cdot \text{ReLU}(W_1 \cdot x_q))$ with $W_1 \in \mathbb{R}^{384\times32}$, $W_2 \in \mathbb{R}^{32\times3}$. This allows different query types to emphasize different geometric components.

3.3 Theoretical Framing

The product manifold $\mathcal{M} = \mathbb{R}^{d_1} \times \mathbb{B}^{d_2} \times \mathbb{S}^{d_3}$ is a Riemannian manifold whose metric tensor is the direct sum of the component metrics. Gu et al. (2019) showed that for graphs with mixed structure—containing both tree-like and cyclic subgraphs—product spaces achieve lower distortion than any single-geometry embedding of the same total dimension. Our hypothesis is that document collections exhibit analogous mixed structure: hierarchical topic taxonomies (favoring $\mathbb{B}$), mutually exclusive document categories (favoring $\mathbb{S}$), and flat within-topic similarity (favoring $\mathbb{R}$).

4 Baseline Setup

BM25. We use the `rank_bm25` Python library with default Okapi BM25 parameters (k_1=1.5, b=0.75). Documents are tokenized with simple whitespace splitting and lowercasing.

Frozen SBERT. We use `all-MiniLM-L6-v2` (384-dimensional) as the frozen encoder. Retrieval is performed via cosine similarity over the full 384d output without any projection. Embeddings are L_2-normalized before computing dot products.

Dataset Splits. For SciFact and FiQA, we use the official BEIR train/test splits and qrels. For our custom Wiki dataset, we use a fixed 80/20 split stratified by query type, ensuring each query type is represented proportionally in both train and test sets.

Retrieval Protocol. All methods retrieve from the full corpus (no pre-filtering or top-k reranking). We compute exact nearest-neighbor search over the entire document pool for each query. Metrics (NDCG@10, Recall@10, Recall@100) are computed using `pytrec_eval` following BEIR conventions.

Dimensionality Note. The trained projection heads use 64 total dimensions by default, compared to 384d for frozen SBERT. This dimensionality reduction introduces a capacity bottleneck: on FiQA, where frozen SBERT already performs well, 64d trained heads cannot match the full 384d frozen baseline. We report frozen SBERT results for reference but emphasize that the primary comparison is *among trained heads at equal dimensionality*, which isolates the effect of geometric inductive bias from capacity differences.

5 Experiments

5.1 Datasets

- **SciFact** (BEIR): 5,183 biomedical abstracts; 809/300 train/test queries. Mildly hierarchical (biomedical taxonomy).
- **FiQA** (BEIR): 57,638 financial documents; 5,500/648 train/test queries. Flat structure.
- **Wiki** (custom): ∼1,000 Wikipedia chunks with 4-level hierarchy across 8 root categories (Biology, Physics, Math, CS, History, Philosophy, Chemistry, Economics). Each document has explicit hierarchy metadata (category → article → section). We generate three types of test queries: *generalize* (section→category, 600 queries), *hierarchy_navigation* (article→category, 200), and *hierarchy_disambiguation* (disambiguate same-titled sections across branches, 165).

5.2 Baselines and Methods

Baselines: (i) BM25 (Okapi, lexical heuristic); (ii) Frozen SBERT (`all-MiniLM-L6-v2`, 384d, cosine similarity).
Proposed methods: Euclidean, Poincaré, and Product projection heads (default 64d total). All retrieve top-100 candidates at test time. Each configuration is run with multiple random seeds; we report mean ± std.

Training Pair Construction. For BEIR datasets (SciFact, FiQA), we construct query–positive pairs from the official qrels: each query is paired with all documents marked relevant (relevance ≥ 1). When a query has multiple relevant documents, we randomly sample one positive per training instance. Hard negatives for BEIR datasets are mined using the frozen SBERT encoder: for each query, we retrieve the top-50 documents by cosine similarity and select the highest-ranked non-relevant documents as hard negatives (4 per query), supplemented by 3 random negatives from the corpus.

For the Wiki dataset, hard negatives are constructed from the hierarchy structure: documents sharing the same section title but belonging to a different top-level category branch serve as hard negatives (e.g., the "History" section under Physics vs. under Economics). This provides 4 hard negatives per query when available, plus 3 random negatives.

SciFact yields ∼920 training triplets, FiQA ∼5,500, and Wiki ∼800. We do not use a separate validation set; training epochs are fixed per head type (see Section 3) without early stopping, as we found performance stable across the last 5 epochs in preliminary runs.

5.3 Main Results

The overall retrieval performance across three datasets is summarized in Table 2.

Table 2: Main retrieval results (64d projection heads, 2 seeds). Best overall in **bold**; best trained projection head underlined.

Dataset	Method	NDCG@10	R@10	R@100
Wiki	BM25	0.079	0.119	0.215
	SBERT (frozen)	0.099	0.155	0.277
	Euclidean-64	**0.578**±.003	0.828±.001	0.967±.001
	Poincaré-64	0.492±.001	0.738±.001	0.936±.002
	Product-64	0.542±.001	**0.839**±.004	**0.985**±.001
SciFact	BM25	0.560	0.686	0.793
	SBERT (frozen)	0.645	0.783	**0.925**
	Euclidean-64	**0.672**±.013	**0.821**±.009	0.922±.005
	Poincaré-64	0.449±.006	0.573±.006	0.797±.002
	Product-64	0.577±.005	0.736±.014	0.903±.009
FiQA	BM25	0.159	0.204	0.359
	SBERT (frozen)	**0.369**	**0.441**	**0.706**
	Euclidean-64	0.238±.002	0.300±.005	0.573±.001
	Poincaré-64	0.183±.001	0.247±.003	0.469±.002
	Product-64	0.263±.003	0.332±.003	0.596±.007

On Wiki and SciFact, all trained heads massively outperform BM25 and frozen SBERT, with Euclidean achieving the highest overall NDCG@10. On FiQA, frozen SBERT (384d) outperforms all 64d trained heads due to the capacity bottleneck, but among trained heads, Product is consistently best. Across all three datasets, Product achieves the highest Recall@100, suggesting that mixed geometry helps surface relevant documents even when ranking precision is lower.

Per-Type Breakdown (Wiki). The aggregate numbers mask an important fine-grained pattern:

Table 3: Wiki NDCG@10 by query type (64d, 2 seeds). Best per type in **bold**.

Method	Generalize	Hier. Disambig.	Hier. Navigation
Euclidean-64	**0.611**$\pm$.006	0.452$\pm$.001	**0.580**$\pm$.002
Poincaré-64	0.551$\pm$.002	0.284$\pm$.003	0.488$\pm$.002
Product-64	0.565$\pm$.001	**0.543**$\pm$.007	0.470$\pm$.003

Product outperforms Euclidean by **+9.1 NDCG@10 points** on hierarchy disambiguation—queries that require distinguishing same-titled sections across different hierarchy branches (e.g., “History” under AI vs. under WWII). This is the task where mixed-curvature geometry has the clearest structural advantage: the spherical component encodes branch membership, and the hyperbolic component encodes depth, jointly enabling disambiguation that flat Euclidean space cannot achieve as effectively.

6 Analysis

6.1 Dimension Sweep

We sweep over embedding dimensions $d \in \{8, 16, 32, 64\}$ across all three methods and datasets (3 seeds each). Table 4 shows NDCG@10.

Table 4: NDCG@10 across embedding dimensions (3 seeds). Best method per cell in **bold**.

Dataset	Method	d=8	d=16	d=32	d=64
Wiki	Euclidean	**.434**$\pm$.005	**.507**$\pm$.004	**.548**$\pm$.001	**.574**$\pm$.009
	Poincaré	.309$\pm$.012	.382$\pm$.003	.431$\pm$.003	.505$\pm$.010
	Product	.360$\pm$.047	.478$\pm$.012	.532$\pm$.007	.535$\pm$.011
SciFact	Euclidean	**.149**$\pm$.008	**.366**$\pm$.005	**.567**$\pm$.016	**.675**$\pm$.012
	Poincaré	.144$\pm$.004	.274$\pm$.010	.379$\pm$.008	.446$\pm$.006
	Product	.100$\pm$.002	.282$\pm$.023	.487$\pm$.002	.589$\pm$.017
FiQA	Euclidean	.039$\pm$.002	**.120**$\pm$.004	.203$\pm$.000	.239$\pm$.002
	Poincaré	**.043**$\pm$.002	.103$\pm$.003	.146$\pm$.002	.183$\pm$.001
	Product	.028$\pm$.005	.108$\pm$.005	**.210**$\pm$.002	**.267**$\pm$.006

Interestingly, the ’low-dim hyperbolic advantage’ observed in our synthetic tree experiments did not directly appear in the real-world datasets. Instead, we observed that Euclidean projections remained dominant on Wiki and SciFact across all tested dimensions. The Product head shows high variance at d=8 ($\pm$.047 on Wiki) because splitting 8 dimensions three ways (4+2+2) leaves too few per component. However, Product overtakes Euclidean on FiQA at $d \geq 32$, suggesting the mixed-curvature approach adds representational capacity that helps on large, flat corpora.

6.2 Component Ablation

We fix total dimension to 64 and systematically remove each geometric component from the Product head (Table 5). Experiments run on Wiki (3 seeds).

Table 5: Component ablation on Wiki (dim=64, 3 seeds). "Disambig." = NDCG@10 on hierarchy disambiguation queries only.

Variant	(E, H, S)	Overall NDCG@10	Disambig. NDCG@10
Euclidean only	(64, 0, 0)	**0.577**$\pm$.016	0.435
Product no-S	(48, 16, 0)	0.544$\pm$.001	0.532
Product no-H	(48, 0, 16)	0.541$\pm$.003	**0.537**
Product full	(32, 16, 16)	0.533$\pm$.006	0.529
Product no-E	(0, 32, 32)	0.523$\pm$.011	0.524
Poincaré only	(0, 64, 0)	0.462$\pm$.005	0.273

Two findings stand out: (1) Removing the Euclidean component hurts most (0.523 vs. 0.533), confirming flat similarity is the dominant signal. (2) All Product variants dramatically outperform Euclidean-only on disambiguation (+8.9–10.2 points), while Euclidean-only is best overall. Interestingly, removing either the hyperbolic or spherical component individually slightly *improves* overall NDCG, suggesting the full three-way product may be over-parameterized at dim=64.

6.3 Adaptive Weight Analysis

The query-adaptive Product head learns per-query weights (α, β, γ) via a small MLP. On Wiki, we analyze the learned weights by query type:

Table 6: Mean learned adaptive weights by QA type (Wiki). Higher = more influence on distance.

QA Type	α (Euclid)	β (Hyper)	γ (Sphere)
Generalize	2.52$\pm$.37	0.47$\pm$.23	2.41$\pm$.32
Hier. disambiguation	3.06$\pm$.32	0.55$\pm$.19	2.73$\pm$.25
Hier. navigation	2.94$\pm$.28	0.45$\pm$.14	2.37$\pm$.22

The hyperbolic weight β is consistently suppressed (0.45–0.55) compared to α and γ (2.4–3.1). Contrary to our hypothesis, hierarchy-related queries do *not* learn higher hyperbolic weights; instead, they increase the Euclidean and spherical weights. This suggests that on our Wiki corpus, hierarchical structure is better captured by the combination of flat similarity and angular separation than by Poincaré distance alone.

The adaptive head improves over the static Product head on SciFact (+1.8 NDCG@10 points, from 0.541 to 0.560) but slightly hurts on Wiki (-1.2) and FiQA (-1.0), indicating the additional parameters overfit on smaller datasets.

6.4 Error Analysis

When does Product fail? Product underperforms Euclidean most on *hierarchy_navigation* queries (0.470 vs. 0.580 NDCG@10). These queries ask "What broader category does article X belong to?" and require matching specific articles to abstract category documents. Euclidean cosine similarity handles this effectively because the query explicitly names the article, providing strong lexical overlap. The Product head's more complex distance metric appears to diffuse the signal.

When does Product excel? Disambiguation queries present same-titled sections from different branches (e.g., "In the context of *AI > Machine Learning*, what does the History section cover?"). BM25 and SBERT achieve 0.449 and 0.548 NDCG@10 respectively by keyword matching, but cannot distinguish which "History" is meant. The Product head reaches 0.543 by leveraging the spherical component to separate branches and the hyperbolic component to encode depth, while Euclidean-only achieves only 0.435.

Qualitative Example. Table 7 shows a representative hierarchy-disambiguation retrieval. The Euclidean head retrieves a same-titled section from the wrong branch, while the Product head correctly identifies the target.

Table 7: Qualitative retrieval example on hierarchy disambiguation.

Query	Euclidean Top-1	Product Top-1
"In the context of AI > Machine Learning, what does the History section cover?"	History section under *Philosophy > Epistemology* (wrong branch)	History section under *CS > Machine Learning* (correct)
"In the context of Biology > Genetics, what does the Applications section cover?"	Applications section under *Chemistry > Organic Chemistry* (wrong branch)	Applications section under *Biology > Genetics* (correct)

7 Limitations and Broader Impact

Our study has several limitations. First, the SBERT encoder is frozen; jointly fine-tuning the encoder with manifold-aware objectives could better align the embedding space with non-Euclidean geometry. Second, our custom Wiki dataset ($\sim$1,000 docs) is small and synthetically constructed—results may not transfer to organic hierarchical corpora at scale. Third, the Poincaré head's consistent underperformance may reflect optimization difficulty (sensitivity to learning rate and tangent-space scale) rather than a fundamental geometric limitation; more extensive hyperparameter search could close the gap.

Fourth, because the Wiki benchmark is synthetically generated from explicit hierarchy metadata, and hard negatives are specifically constructed around hierarchy structure, it may overestimate the benefit of geometry-aware retrieval on naturally occurring user queries.

Fifth, some gains of the Product head may arise from architectural factorization (multiple parallel projection heads, spherical normalization, extra learned weight parameters) rather than from curvature itself. Future work should compare against Euclidean multi-head baselines with matched parameterization to isolate the contribution of non-Euclidean geometry.

Broader Impact. Geometry-aware retrieval could improve RAG systems in domains with rich hierarchical structure (medicine, law, scientific taxonomies) by making retrieval more structure-aware, potentially reducing hallucination from retrieving superficially similar but contextually wrong documents. However, structure-aware retrieval can also amplify errors in an incorrect taxonomy: if the hierarchy metadata is biased or outdated, the retriever may systematically prefer documents from the wrong branch, compounding rather than correcting organizational biases.

8 Conclusion

We presented Manifold-RAG, a framework for dense retrieval in mixed-curvature product manifolds combining Euclidean, hyperbolic, and spherical geometry. Through experiments on three datasets with dimension sweeps, component ablations, and adaptive weight analysis, we find that while Euclidean projections achieve the best overall retrieval quality, the product manifold provides a clear advantage on structure-sensitive tasks—particularly hierarchy disambiguation (+9.1 NDCG@10 over Euclidean) and FiQA retrieval at reduced dimensions. The adaptive weight analysis reveals that the model learns to suppress the hyperbolic component in favor of Euclidean and spherical distances, suggesting that the practical value of mixed-curvature geometry in retrieval lies more in the *combination* of flat and angular similarity than in hyperbolic distance alone.

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A ADDITIONAL EXPERIMENT DETAILS

Hyperparameters. Batch size: 256 for all heads. Gradient clipping: max_norm=1.0. Negatives per sample: 7 (4 hard + 3 random when hard negative index is available). Margin: 0.5. Hierarchy norm regularization: $\lambda = 0.2$, targets $\{0.20, 0.45, 0.70\}$ for levels $\{1, 2, \geq 3\}$. SBERT embeddings are cached as `.npy` files to avoid redundant encoding across experiments.

Dimension Allocation Ablation. We test five allocation strategies for the Product head at total dim=64:

On Wiki, allocations are within noise (± 0.003). On SciFact and FiQA, Euclidean-heavy (48,8,8) is best, consistent with the component ablation finding that flat similarity is the dominant signal on non-hierarchical data. The hyperbolic-heavy split (16,32,16) notably hurts SciFact (-6.2 points vs. default).

Table 8: NDCG@10 for different (E,H,S) allocations at total dim=64 (3 seeds).

Allocation (E,H,S)	Wiki	SciFact	FiQA
(48, 8, 8) Euclid-heavy	.536±.004	**.574**±.007	**.276**±.008
(32, 16, 16) Default	.533±.006	.541±.013	.273±.004
(22, 21, 21) Equal	.536±.008	.558±.014	.274±.001
(16, 16, 32) Sphere-heavy	.534±.004	.553±.006	.273±.009
(16, 32, 16) Hyper-heavy	.533±.009	.479±.013	.270±.003

Wiki QA Generation. Disambiguation queries are generated programmatically: for each section title appearing under $\geq$2 different top-level categories, we create a question embedding the hierarchy path (e.g., "In the context of 'AI > Machine Learning', what does the 'History' section cover?"). The ground-truth document is the specific section chunk; same-titled chunks from other branches serve as hard negatives during training. Navigation queries ask "What broader category does article X belong to?" with the category document as ground truth. Generalize queries present a snippet from a section (with category/title keywords replaced by "[topic]") and ask which broad subject area it belongs to.

Hardware. All experiments were run on a single NVIDIA RTX 4060 Laptop GPU (8GB VRAM). Total GPU time: approximately 8 hours for all experiments including dimension sweep, ablations, and adaptive head training.

Software. Python 3.10, PyTorch 2.1, `sentence-transformers` 2.2, `geoopt` 0.5 (for Riemannian optimization and Poincaré ball operations), `rank_bm25` 0.2 (for BM25 baseline), and `pytrec_eval` 0.5 (for NDCG/Recall computation following TREC conventions).

Random Seeds. Main results (Table 3) use 2 seeds; dimension sweep and ablation experiments (Tables 5–7) use 3 seeds. We report mean $\pm$ standard deviation across seeds. All seeds control random negative sampling, weight initialization, and training data shuffling.